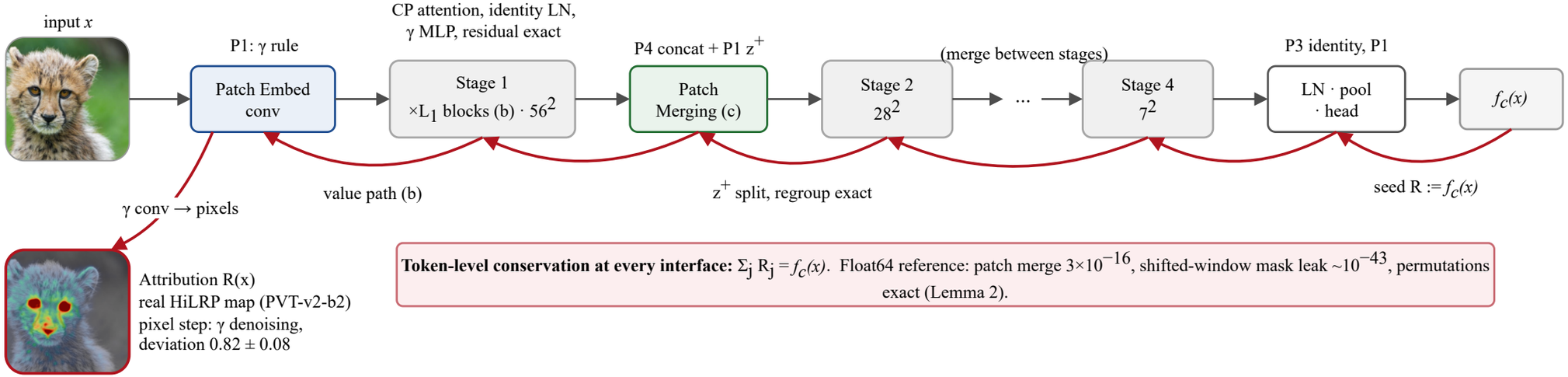
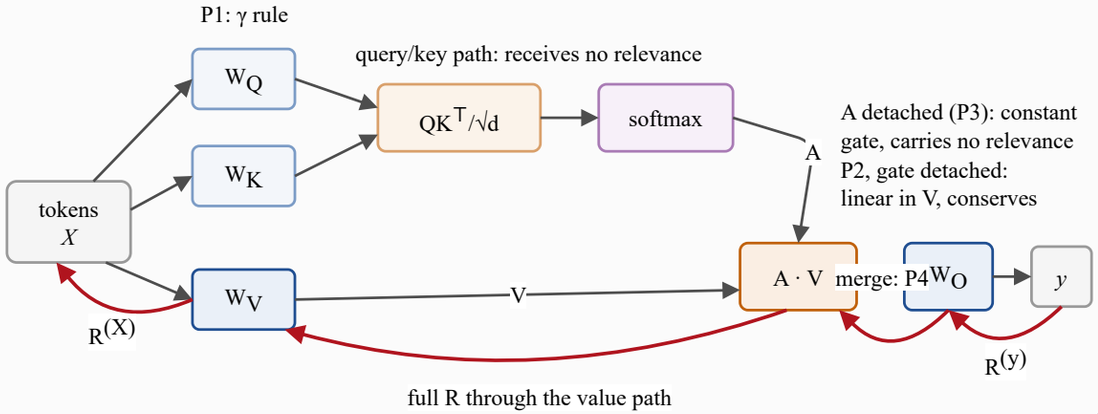
(a) HiLRP: one conservation-valid backward pass over any hierarchical or hybrid ViT

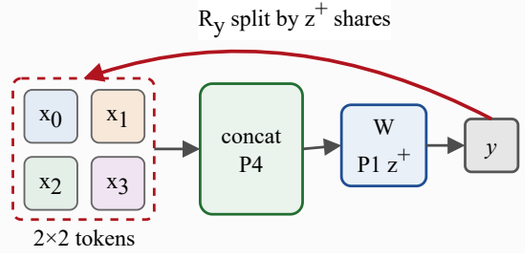


Token-level conservation at every interface: $\Sigma_j R_j = f_c(x)$. Float64 reference: patch merge 3×10^{-16}, shifted-window mask leak $\sim10^{-43}$, permutations exact (Lemma 2).

(b) One block: CP-LRP attention, relevance takes the value path



(c) Patch merging: exact rule



$$R_{x_k} = \Sigma_c \; x_k W^+_{k,c} / (\Sigma_{k'} x_{k'} W^+_{k',c} + \varepsilon) \cdot R_{y,c}$$

conserves to 3×10^{-16}; un-concat regroup and cyclic shift are permutations: exact, no leak

(d) Four conserving primitives

- **P1 linear**: Q/K/V/out, MLP, conv, pool → ε/γ rule
- **P2 bilinear**: QK^T, A·V, linear attn. → split, or CP: detach one factor
- **P3 normalize/gate**: softmax, LN, denominators → detach → identity
- **P4 reindex**: heads, windows, shifts, merge → permutation, exact

A new ViT is a new arrangement of the same four: covered by construction, no new derivation.